



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



CPDLC over ATN-B1 Aviation Link Vulnerability Requires Safety-Critical Exposure Review

Threat Report

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TLP:CLEAR | Lost Boys Cyber | CPDLC over ATN-B1 Aviation Link Vulnerability Requires Safety-Critical Exposure Review - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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1. Executive Summary

BLUF: CISA ICS advisory coverage of CPDLC/ATN-B1 weaknesses highlights aviation communications availability and workload risk from malformed or out-of-sequence frames. This report is a finished Lost Boys Cyber daily intelligence product for 2026-09-14. Source recency was prioritized for the last 24-72 hours and expanded to 7 days or standing sector references where necessary for corroboration.

Field	Value
Daily Section	Top 5 aviation/aerospace threats
Severity	High
Confidence	High
Primary Exposure	Air traffic communications, avionics integration, airline flight operations, and aviation safety engineering.
Analyst Decision	Review controls, hunt telemetry, and update watchlists/playbooks

2. Intelligence Requirements

Requirement	Answer
What changed?	CISA ICS advisory coverage of CPDLC/ATN-B1 weaknesses highlights aviation communications availability and workload risk from malformed or out-of-sequence frames.
Who should care?	Air traffic communications, avionics integration, airline flight operations, and aviation safety engineering.
What should analysts do today?	Validate exposure, run hunts, and review third-party/control exceptions

3. Source Review and Confidence

#	Source	URL	Role
1	CISA ICSA-26-219-01 CPDLC over ATN-B1 Vulnerabilities	https://www.cisa.gov/news-events/ics-advisories/icsa-26-219-01	Used for corroboration and defender context
2	CISA ICS Advisories index	https://www.cisa.gov/news-events/ics-advisories	Used for corroboration and defender context
3	EASA cybersecurity domain	https://www.easa.europa.eu/en/domains/cybersecurity	Used for corroboration and defender context

Confidence is High. This product avoids unverified attribution and treats vendor/news reporting as operational cues pending internal validation.

4. Threat / Campaign Overview

The selected reporting indicates a defender-relevant pressure point: CISA ICS advisory coverage of CPDLC/ATN-B1 weaknesses highlights aviation communications availability and workload risk from malformed or out-of-sequence frames. The likely operational pattern is not a single IOC-only event; it is a class of behavior that combines exposure management, identity control, endpoint visibility, and rapid triage.

Signal	Why It Matters	Collection
CPDLC	Primary analytic pivot for this topic	SIEM/EDR/vulnerability/source logs
ATN-B1	Secondary behavior or exposure clue	Identity, proxy, EDR, cloud logs
aviation VHF datalink	Triage and watchlist value	Case notes, watchlists, alert enrichment

5. Tactics, Techniques, and Procedures

Tactic	Technique / Behavior	Defensive Note
Initial Access	Phishing, exploit delivery, exposed remote access, or supplier access	Prioritize high-value users/assets and vendor paths
Execution	Scripted execution, payload staging, or malicious automation	Monitor parent/child process chains and command-line anomalies
Credential Access	Token, password, or session theft where relevant	Review impossible travel, risky sign-ins, OAuth app grants
Command and Control	Reused infrastructure, cloud services, or remote-management tooling	Enrich DNS/proxy/EDR network events and block known-bad only after validation

6. Infrastructure, Indicators, and Observables

Observable	Type	Action
CPDLC	Observable / analytic pivot	Validate in SIEM, EDR, vulnerability, and ticketing systems
ATN-B1	Observable / analytic pivot	Validate in SIEM, EDR, vulnerability, and ticketing systems
aviation VHF datalink	Observable / analytic pivot	Validate in SIEM, EDR, vulnerability, and ticketing systems

No hard atomic IOCs are asserted unless present in the cited public sources. Treat these entries as observables and hunting pivots.

7. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Likely Exposure Path	Defender Action
Airlines / operators	Disruption to passenger, crew, dispatch, or safety-adjacent support workflows	Identity, SaaS, endpoint, and third-party access	Validate MFA, conditional access, and offline continuity playbooks

Airports / ground services	Shared IT and vendor concentration can cascade across tenants	Managed service provider, network segmentation, remote access	Review vendor access and tabletop outage procedures
Aerospace / MRO / space	Engineering data, maintenance records, telemetry pipelines, and supplier portals are high-value	Cloud repositories, remote administration, supplier portals	Enforce least privilege, backup restore tests, and supplier assurance

8. Detection Engineering

<pre>DeviceProcessEvents where Timestamp > ago(7d) where ProcessCommandLine has_any ("powershell", "rundll32", "mshta", "regsvr32", "curl", "bitsadmin", "rmm", "screenconnect", "anydesk") summarize Events=count(), Hosts=dcount(DeviceName), Samples=make_set(ProcessCommandLine, 10) by FileName, InitiatingProcessFileName order by Events desc</pre>
<pre>SigninLogs where TimeGenerated > ago(7d) where ResultType != 0 or RiskLevelDuringSignIn in ("medium", "high") summarize Failures=count(), Users=dcount(UserPrincipalName), IPs=make_set(IPAddress, 20) by AppDisplayName, ConditionalAccessStatus order by Failures desc</pre>
<pre>title: Daily Threat Review Suspicious Remote Tool Or Script Chain status: experimental logsource: product: windows category: process_creation detection: selection_tools: CommandLine contains: - 'screenconnect' - 'anydesk' - 'teamviewer' - 'powershell -enc' - 'rundll32' condition: selection_tools fields: - UtcTime - Image - ParentImage - CommandLine - User level: medium</pre>

9. Threat Hunting

<pre>DeviceNetworkEvents where Timestamp > ago(14d) where RemoteUrl has_any ("pastebin", "githubusercontent", "workers.dev", "ngrok", "trycloudflare", "duckdns") summarize Events=count(), Hosts=dcount(DeviceName), Urls=make_set(RemoteUrl, 25) by InitiatingProcessFileName order by Events desc</pre>
<pre>index=endpoint earliest=-14d (process_name IN ("powershell.exe", "rundll32.exe", "mshta.exe", "regsvr32.exe") OR command_line="*screenconnect*" OR command_line="*anydesk*") stats count dc(host) as hosts values(command_line) as sample_commands by parent_process_name process_name user sort - count</pre>

Hunt intent: identify behavior classes related to this daily item, then pivot through affected business units and source-specific observables.

10. Risk Assessment

Risk	Rating	Rationale
Operational disruption	Medium-High	Dependent services, identity, or endpoints can affect business continuity
Data exposure	Medium	Credential/session compromise and supplier paths can expose sensitive data
Detection gap	Medium	Many signals require cross-domain telemetry correlation

11. Recommended Actions

Priority	Action	Owner
P1	Run the included hunts against the last 7-14 days	SOC / Detection Engineering
P2	Validate control exceptions and internet/vendor exposure	Vulnerability / IT Ops
P3	Add this topic to daily watchlists and case-triage notes	CTI / SOC Lead

12. References

- [1] CISA ICSA-26-219-01 CPDLC over ATN-B1 Vulnerabilities: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-219-01>
- [2] CISA ICS Advisories index: <https://www.cisa.gov/news-events/ics-advisories>
- [3] EASA cybersecurity domain: <https://www.easa.europa.eu/en/domains/cybersecurity>